

Supplementary Material for “Target–Deployment Routed Refinement”

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Proofs and Additional Analysis

Behavior-anchor decomposition and Proposition 1

For any square-integrable random action a conditioned on s , let $b(s) = \mathbb{E}[a \mid s]$. The bias–variance identity gives

$$\mathbb{E}[\|\mu(s) - a\|_2^2 \mid s] = \|\mu(s) - b(s)\|_2^2 + \mathbb{E}[\|a - b(s)\|_2^2 \mid s]. \quad (1)$$

Averaging over states and dividing by d proves Eq. (7); the second term is independent of μ .

For state-conditioned deterministic policies $\pi_\mu(\cdot \mid s) = \delta_{\mu(s)}$ and $\pi_\nu(\cdot \mid s) = \delta_{\nu(s)}$,

$$W_2^2(\pi_\mu(\cdot \mid s), \pi_\nu(\cdot \mid s)) = \|\mu(s) - \nu(s)\|_2^2. \quad (2)$$

Consequently, d_d^2 is a fixed rescaling of the integrated squared Wasserstein distance. The normalized anchored objective is

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + d_d^2(\mu, \nu). \quad (3)$$

Multiplication by the positive constant $1/\alpha$ identifies it with Eq. (8) when $1/(2h) = 1/\alpha$, or $h = \alpha/2$. If the value term is instead $-\lambda \mathbb{E}_s \hat{Q}$, then $\lambda \hat{Q} = \lambda C \bar{Q}$ and $h = \lambda C/2$.

Proof of Proposition 2

By ε_k -optimality relative to the feasible comparator μ_{k-1} ,

$$\begin{aligned} & -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_k(s)) + d_d^2(\mu_k, \mu_{k-1}) \\ & \leq -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_{k-1}(s)) + \varepsilon_k. \end{aligned} \quad (4)$$

Rearranging yields Eq. (13).

Critic-error transfer

Let $\hat{Q} = Q^\dagger + e$. Substitution into Eq. (13) gives

$$\begin{aligned} m_k^{\hat{Q}} &= \frac{2h_k}{C_k} \mathbb{E}_s [Q^\dagger(\mu_k) - Q^\dagger(\mu_{k-1})] - d_d^2(\mu_k, \mu_{k-1}) \\ &+ \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})] \\ &= m_k^{Q^\dagger} + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})]. \end{aligned} \quad (5)$$

This proves Eq. (14). If $|e(s, a)| \leq \delta_k$ on the realized path, then

$$m_k^{Q^\dagger} \geq -\varepsilon_k - \frac{4h_k}{C_k} \delta_k. \quad (6)$$

The result concerns a trusted critic objective. A true-return statement would require additional occupancy and value-estimation assumptions.

Proof of Proposition 3

Let $F_{s'}(a) = \min_j Q_j^-(s', a)$. If every $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz, then

$$\begin{aligned} |F_{s'}(a) - F_{s'}(b)| &\leq \max_j |Q_j^-(s', a) - Q_j^-(s', b)| \\ &\leq L(s') \|a - b\|_2. \end{aligned} \quad (7)$$

Therefore,

$$|y_{\mu_1} - y_{\mu_0}| \leq \gamma L(s') \|\mu_1(s') - \mu_0(s')\|_2. \quad (8)$$

Squaring, averaging over dataset transitions, and taking the square root proves Eq. (15). Later actors do not occur because the TD target conditions only on μ_1 .

Local explicit–implicit equivalence

Let $\beta = dh/C$, $g = \nabla_a \hat{Q}(a_0)$, and suppose the action constraint is inactive. The scale-matched linearized update is $\Delta a_{\text{lin}} = \beta g$. A statewise exact proximal solution satisfies

$$\Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (9)$$

If $\hat{Q}$ is twice continuously differentiable near a_0 ,

$$\nabla_a \hat{Q}(a_0 + \Delta a) = g + H \Delta a + O(\|\Delta a\|^2). \quad (10)$$

Thus

$$(I - \beta H) \Delta a_{\text{prox}} = \beta g + O(\beta \|\Delta a_{\text{prox}}\|^2). \quad (11)$$

For $\beta \|H\| < 1$ and sufficiently small β ,

$$\Delta a_{\text{prox}} = \beta g + \beta^2 H g + O(\beta^3), \quad (12)$$

whereas $\Delta a_{\text{lin}} = \beta g$. Both are first-order schemes for fixed positive C , and changing a fixed C only rescales time. A positive policy-dependent global C preserves the raw-critic ascent direction under an adaptive time parameterization in the infinitesimal limit. This derivation is not applied to the large-budget regime with active projection, ReLU-region crossings, critic drift, or inaccurate actor solves.

Proof of Proposition 4

Write $x_j = x_0 + jhf_0 + O(h^2)$ for fixed K , which follows inductively from the one-step consistency assumption. Smoothness then gives

$$f(x_j) = f_0 + jhJ_0f_0 + O(h^2). \quad (13)$$

Summing the K increments,

$$\begin{aligned} x_K - x_0 &= h \sum_{j=0}^{K-1} f(x_j) + h^2 \sum_{j=0}^{K-1} p(x_j) + O(Kh^3) \\ &= Khf_0 + h^2 \frac{K(K-1)}{2} J_0f_0 \\ &\quad + Kh^2 p_0 + O(h^3). \end{aligned} \quad (14)$$

where the final remainder uses fixed K . Substituting $h = T/K$ proves Eq. (18). The exact flow satisfies $x(T) = x_0 + Tf_0 + \frac{1}{2}T^2J_0f_0 + O(T^3)$. Thus explicit Euler has leading error $-T^2J_0f_0/(2K)$, while backward Euler has $+T^2J_0f_0/(2K)$. Both remain first-order methods.

Experimental Details

Algorithms and schedules

For the fixed-budget TD3+BC study, the total normalized budget is split as $h = T/K$. TDRR-Prox uses Eq. (9) with a moving reference; TDRR-Lin constructs Eq. (19) and uses MSE regression. The K actor networks share an architecture and learning rate but are independently initialized, persistent networks with separate Adam states. At every second critic update, the implementation performs one optimizer step per actor in chain order on the same minibatch. The base actor is updated first; then its actor target and the critic target receive a Polyak update of 0.005 before later actors use stop-gradient references from the updated online chain. The scale C_k is recomputed from each hop’s reference actions. All methods therefore match data, critic architecture, and 10^6 critic updates, but a K -hop run performs K actor optimizer steps and is not compute matched. We use batch size 256, learning rate 3×10^{-4} , discount 0.99, and ten evaluation episodes every 5000 updates. The 14-value grid spans 0.05 to 20; the complete sweep uses two paired seeds and reports both cell-level and raw-seed failure counts.

Diagnostic quantities

On a fixed current-state validation batch with paired dataset actions, we compute the empirical action-averaged proxies

$$\begin{aligned} \hat{D}_{\text{critic}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_1(s) - a\|_2^2, \\ \hat{D}_{\text{final}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_K(s) - a\|_2^2. \end{aligned}$$

These quantities measure the online actor chain relative to its sample anchor. They are proxies for, rather than direct measurements of, the next-state target-policy

displacement in Proposition 3: the practical TD target uses a Polyak target actor and policy-smoothing noise. By Eq. (7), the population expectation of each proxy equals squared displacement from the conditional-mean behavior map plus the same additive conditional-action variance. Population differences therefore cancel this common floor, but ratios do not: a ratio of sample-anchored proxies is not a ratio of mean-policy displacements. Finite validation batches add sampling error to both interpretations. We additionally evaluate the deterministic Polyak target actor and 64 draws of clipped target-smoothing noise on 4096 next-state transitions per checkpoint. All target-exposure quantities remain squared displacement per action coordinate relative to the paired next dataset action. Thus absolute levels include conditional action variance, whereas paired differences use a common sampled anchor. For TDRR-Lin, projection attenuation is the RMS norm ratio between the clipped and raw target displacements. If $\tilde{g}$ is the normalized critic action gradient and Δa is the learned actor displacement, define

$$h_{\text{eff}} := \frac{\mathbb{E} \langle \Delta a, \tilde{g} \rangle}{\mathbb{E} \|\tilde{g}\|_2^2}.$$

We report $h_{\text{eff}}/(dT/K)$ as the effective fraction of the nominal linear-target coefficient. The matched audit evaluates 512 fixed states per environment–seed run. We also record action saturation, TD-error quantiles, critic magnitude, critic loss, and gains under both the jointly trained critic and a controlled common critic: the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain is the sum of mean Q changes across the K hops on this shared batch. The common critic supports cross-critic comparison but is not treated as ground truth. A run is descriptively labeled collapsed only from its external normalized return, using the threshold 20.

For the exploratory simulator-reference audit, we clone MuJoCo integration state and environment RNG from evaluation-policy trajectories, record the episode clock, and reset the TimeLimit for a continuing-task critic-calibration estimand. At each state we compare $a_1 = \mu_1(s)$ and $a_K = \mu_K(s)$ under continuations ρ_1 , μ_1 , and μ_K ; the primary calibration target here is the discounted return under the effective backup continuation ρ_1 . The CPU screening covers three environments, $T \in \{4, 7, 10, 14, 20\}$, $K \in \{2, 3\}$, two seeds, and 12 states per checkpoint. It uses a 1000-step horizon, $\gamma = .99$, four common-random-number target-smoothing rollouts, and a one-return-unit ranking dead zone. It is a post-hoc screening study, not a causal routing intervention.

The route-only intervention trains two actor-detached shadow critics from identical initial parameters on the same minibatches and target-smoothing draws while the primary first-hop critic continues to drive the shared actor trajectory. The four environment–budget cells were selected post hoc from visible stability-frontier transitions, so inference is de-

scriptive of this targeted sample. One shadow uses the first-hop target actor and the other the final-hop target actor. The simulator audit evaluates 12 cloned states at four checkpoints in each of eight runs; inference aggregates checkpoint-level state medians to the training-run level.

The compute-matched study uses the same eight final cells. Sequential $K = 3$ performs actor updates anchored at μ_0, μ_1 , and μ_2 ; the fixed-reference control anchors both later actors at μ_1 ; the direct-final control applies two optimizer calls to one $2T/3$ objective anchored at μ_1 . Every condition performs three actor optimizer calls per delayed update. The target-lag control instead keeps $K = 1$ and changes only the actor-target Polyak rate from .005 to .005/3, leaving the critic-target rate unchanged.

The frozen-critic small-step audit uses the final TD3+BC $T = 1$ critic for all nine environments and two seeds. At 512 interior dataset actions per run it compares the projected explicit step with a projected fixed-point solution of the implicit action equation for $h \in \{.001, .003, .01, .03, .1\}$. This is an action-level local-consistency test, not a return or coupled-training experiment.

Additional Results

The paired route-only study uses the same eight runs but aggregates four checkpoints per run. For symmetric relative error against the common first-route simulator estimand, all eight run-level final-route minus first-route differences are positive. Their mean is .124, median is .013, exact two-sided sign-flip $p = .0078$, and run-bootstrap mean 95% interval is [.008, .323]. Because the routes imply different continuation policies, this estimates their relative alignment with the primary first-route value estimand rather than calibrating each shadow to its own continuation; it is not a return-effect estimate.

In the frozen-critic audit, 15 of 18 environment-seed runs have at least one converged, projection-free point eligible for the local comparison. Every eligible run has maximum local relative explicit-implicit discrepancy below 2.7%; the median run maximum is .55%. Three runs have no eligible point, and a log-log slope is identifiable in only one run because a frozen ReLU critic can produce floating-point agreement inside one activation region. The audit supports local numerical agreement but does not estimate a universal $O(h^2)$ rate.

Direct target exposure and simulator-reference screening

The direct target-policy audit covers 270 checkpoints: three methods, nine environments, five budgets, and two seeds. Table 8 separates critic-target exposure from final-policy reach. The ratios are descriptive ratios of sample-anchored squared displacements and are not ratios of conditional-mean policy distances.

The current-state proxy and deterministic target exposure have Pearson correlation .9999 over all 270 cells; their median ratio is .997 and the largest cellwise relative difference is 5.7%. The corresponding correlation after target smoothing is .9993. The close alignment is an empirical property of these checkpoints, where Polyak lag is small, rather than a mathematical equivalence. Among the 84 Prox-3/Prox-2 runs in which Prox-3 lowers deterministic target exposure, Prox-3 is nevertheless worse by more than one score point in 17. This rejects the stronger interpretation that minimizing exposure alone determines performance.

In the simulator-reference screening, 55 stable cells contribute 660 states and five collapsed cells contribute 60. At a_K , the median signed online critic error relative to $Q_{MC}^{p_1}$ is -38.7 in stable cells and $+1.70 \times 10^{12}$ in collapsed cells. This independently corroborates the overestimation signature in Figure 3. We do not treat the sparse subset as a population action-ranking estimate: only 69 of 720 states have both action differences outside the dead zone, with 29 sign disagreements, and within-checkpoint states are correlated.

T	.05	.1	.2	.4	.7	1.5	2.5	4	7	10	12	14	17	20
TD3+BC	65.19	62.81	64.85	64.22	69.14	75.43	76.80	49.16	30.35	27.14	22.08	18.51	15.35	19.21
Prox-2	61.11	66.50	68.87	70.11	74.69	78.52	81.57	72.13	64.57	44.67	30.43	30.73	24.01	27.61
Prox-3	58.31	60.69	68.61	71.32	75.25	78.21	81.55	80.91	70.84	53.91	52.60	41.47	29.23	40.51
Lin-2	60.98	63.12	66.20	72.07	67.98	72.70	80.10	73.95	45.63	39.98	27.09	25.10	24.41	20.63
Lin-3	58.32	59.97	63.69	70.52	73.23	77.14	76.94	75.31	66.47	45.78	30.65	39.54	25.74	23.88

Table 1: Mean score over nine tasks at each total budget in the fixed-budget sweep.

Threshold	TD3	Prox-2	Prox-3	Lin-2	Lin-3
0	2/12	2/6	1/6	0/6	2/12
10	29/66	20/51	15/39	28/54	21/46
20	35/68	25/53	17/41	29/58	24/53
30	36/74	26/56	19/44	29/63	24/55
40	39/83	26/60	20/48	32/64	24/57

Table 2: Collapse-count sensitivity for $T \geq 4$. Entries are seed-mean cells (out of 63) / raw seeded runs (out of 126). Prox-3 has the fewest failures for thresholds 10–40; Lin-3 improves on Lin-2 throughout this range. Threshold 0 isolates negative scores.

Environment	TD3	Prox-2	Prox-3	Lin-2	Lin-3
Hopper-M	16.0	35.2	69.4	25.5	46.3
Hopper-MR	41.4	90.6	99.7	90.5	99.4
Hopper-E	3.4	17.5	30.7	16.6	27.7
HalfCheetah-M	59.0	59.1	59.2	53.4	55.8
HalfCheetah-MR	51.7	52.6	51.6	46.5	49.2
HalfCheetah-E	18.4	24.5	28.5	18.8	22.4
Walker2d-M	10.3	24.3	34.3	9.3	26.5
Walker2d-MR	31.3	49.0	76.6	51.2	43.5
Walker2d-E	2.2	25.4	25.0	18.5	24.4

Table 3: Per-environment high-budget ($T \geq 4$) means.

	Lin st.	Lin col.	Prox st.	Prox col.
No. runs	52	38	59	31
TD p99	270	9.6×10^{23}	160.8	5.0×10^{21}
$ Q $	311	8.5×10^{11}	346.7	6.9×10^{10}
Critic loss	34	1.2×10^{23}	18.7	8.8×10^{20}
$\hat{D}_{\text{critic}}$	.112	.538	.064	.446
$\hat{D}_{\text{final}}$	.147	.626	.094	.578
Common- Q gain	+2.11	−1.42	+2.51	−1.06

Table 4: Matched $K = 2$ diagnostic medians over $T \in \{4, 7, 10, 14, 20\}$.

T	Score		$K = 2$			$K = 3$		
	Lin-2	Lin-3	$\hat{D}_c$	$\hat{D}_f$	R_{rms}	$\hat{D}_c$	$\hat{D}_f$	R_{rms}
4	53.45	64.66	.055	.059	1.040	.054	.062	1.063
7	63.36	74.26	.056	.065	1.075	.055	.069	1.115
10	58.12	85.35	.058	.072	1.113	.056	.077	1.168
12	0.66	50.53	.770	.971	1.123	.352	.475	1.168
14	0.66	41.20	.841	1.025	1.105	.125	.318	1.453
17	0.69	7.73	.786	.935	1.091	.257	.557	1.496
20	1.28	0.66	.854	.952	1.055	.746	1.007	1.162

Table 5: Targeted Hopper-medium comparison of scale-matched linearized $K = 2$ and $K = 3$. The displacement columns are medians over two seeds; R_{rms} is the median of the per-seed RMS ratio $\sqrt{\hat{D}_f/\hat{D}_c}$. Because both sample-anchored proxies contain an additive behavior-action variance floor, R_{rms} is descriptive and is not a population ratio of final to critic-coupled mean-policy displacement.

T	TD3+BC	Prox-2	Prox-3	Prox-4
7	48.1	83.2	94.4	84.8 $\pm$ 8.0
10	0.7	50.8	99.5	100.4 $\pm$ 1.1
12	0.9	0.7	93.6	101.2 $\pm$ 0.4
14	0.7	50.6	88.1	101.1 $\pm$ 0.7
17	0.7	0.6	0.7	100.9 $\pm$ 1.7
20	0.7	0.7	51.5	41.8 $\pm$ 44.0

Table 6: Targeted Hopper-medium scores. The $K \leq 3$ entries are two-seed means from the complete sweep; Prox-4 reports four-seed mean $\pm$ sample standard deviation from a separate-machine run and is used only for the stability-window comparison.

Condition	Mean	Median	Score < 20
Sequential $K = 3$	71.4	88.9	2/8
Fixed-ref. $K = 3$	70.0	87.9	2/8
Direct $K = 3$	59.4	89.6	3/8
Standard $K = 1$	13.6	0.8	6/8
3 $\times$ target lag, $K = 1$	12.0	1.5	7/8

Table 7: Targeted final-return controls on the same eight environment–budget–seed cells. The three $K = 3$ rows are compute matched to three actor optimizer calls; the $K = 1$ rows differ only in actor-target Polyak rate.

Comparison	Quantity	Lower	Ratio
P3 / TD3	Target disp.	88/90	.658
P3 / TD3	Final disp.	36/90	1.057
P3 / P2	Target disp.	84/90	.903

Table 8: Paired CPU audit of deterministic next-state Polyak-target exposure and current-state final-policy displacement. The Lower column counts runs in which the left method has the smaller quantity.

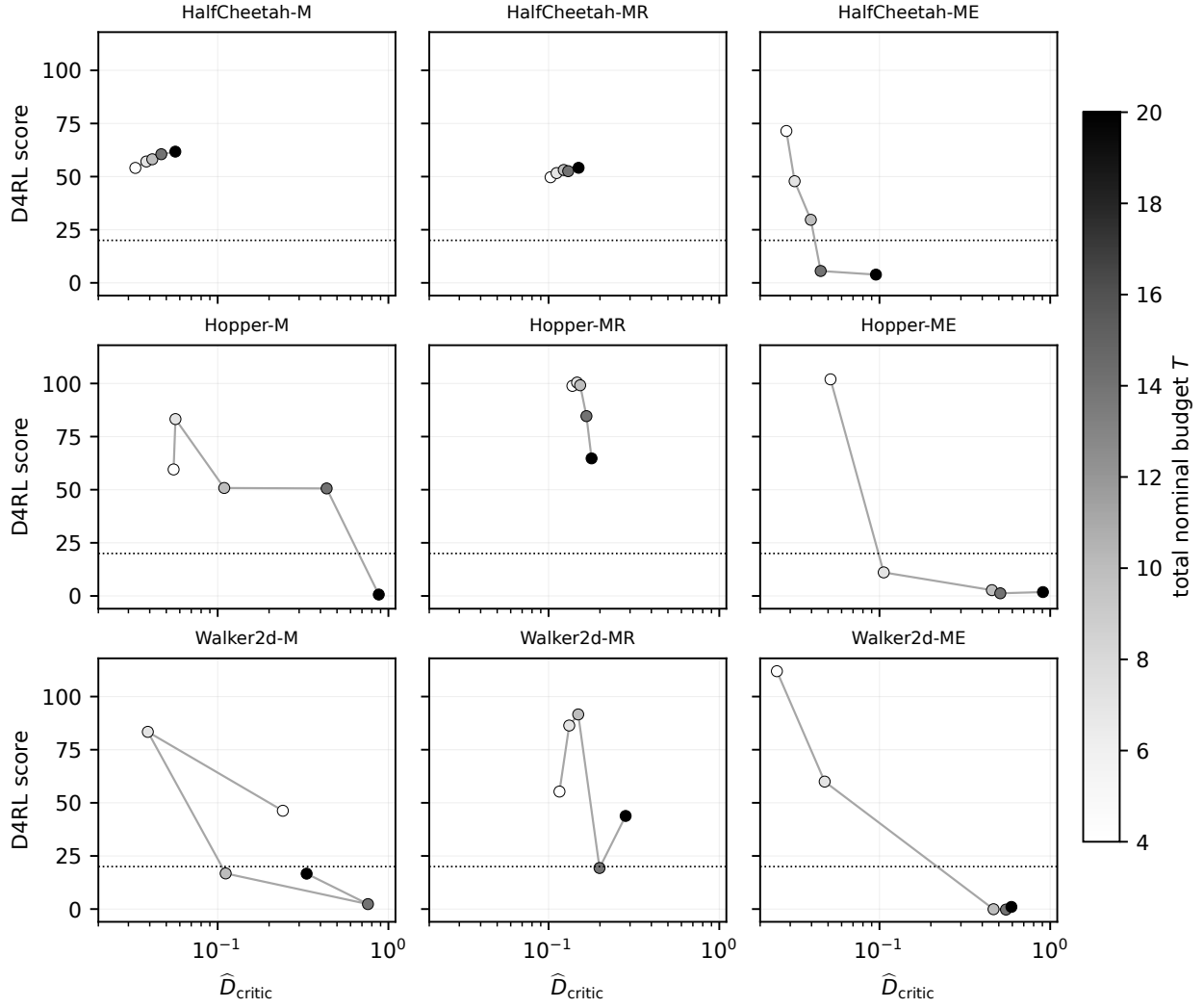


Figure 1: Environment-wise return versus the current-state $\hat{D}_{\text{critic}}$ proxy for TDRR-Prox $K = 2$. Points are two-seed means for the common-batch audit at $T \in \{4, 7, 10, 14, 20\}$; shade denotes T . Lines connect increasing T , exposing reversals rather than imposing a monotone response. Shared axes show that realized movement organizes several within-task transitions but does not define a universal cross-task failure threshold. The proxy does not measure the next-state Polyak target-policy movement.

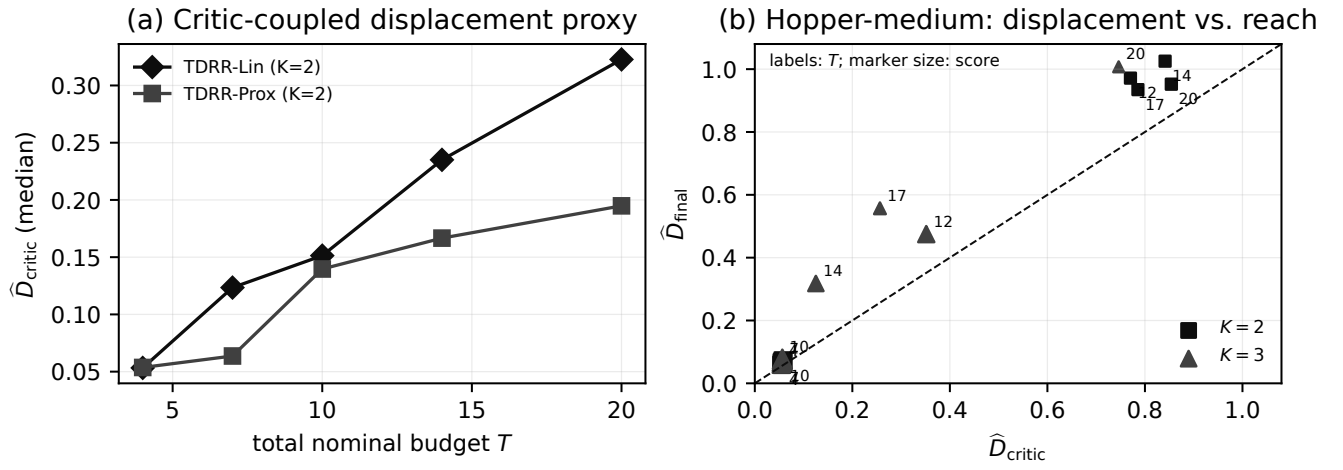


Figure 2: Realized movement. (a) The proximal realization produces a smaller current-state first-hop displacement proxy for $T \geq 7$. (b) On Hopper-medium, $K = 3$ either increases final reach at approximately fixed displacement or contracts both displacements in the observed score-transition region. The displayed current-state quantities exclude Polyak target lag and smoothing noise. Labels indicate T and marker size is proportional to score.